

Central Road Renovation Project

Willow Creek — Project file | Start date: September 1, 2023 | Budget: \$750,000

Project Overview

The central road renovation is a major infrastructure project for Willow Creek. It aims to transform the appearance and functionality of the town center. With an allocated budget of \$750,000, the project was launched on September 1, 2023 to address mobility, accessibility, and public-space quality challenges in the downtown core.

Context and Rationale

Willow Creek's downtown, with its historic streets and aging infrastructure, is facing several urgent issues:

- damaged pavement with numerous potholes and cracks;
- sidewalks that are not fully accessible to people with reduced mobility;
- outdated and insufficient signage;
- not enough dedicated space for active mobility;
- a failing stormwater drainage network;
- aging street furniture that lacks visual consistency.

This renovation is part of a broader downtown revitalization strategy, connected to other projects such as Market Square redevelopment and bike-lane development. It will significantly improve residents' quality of life and support the town's commercial and visitor appeal.

Project Objectives

- 1 Improve road and pedestrian safety through a complete redesign of the roadway and circulation areas.
- 2 Support universal accessibility by removing physical barriers and adapting public infrastructure.
- 3 Promote active mobility by adding bike facilities and shared streets.
- 4 Enhance public space with quality materials and coherent urban design.
- 5 Modernize underground networks to prevent infiltration and flooding.
- 6 Support the local economy by creating an environment that encourages visits to downtown businesses.
- 7 Reduce environmental impact through durable materials and the creation of cooling green spaces.

Scope of Work

The project covers the following corridors and public spaces:

- Republic Avenue — 600 meters / about 0.37 miles;
- Victor Hugo Street — 350 meters / about 0.22 miles;
- City Hall Square and surrounding areas — 2,800 m² / about 30,100 sq ft;
- Commerce Street — 250 meters / about 0.16 miles;
- the main intersection with School Street.

In total, about 1.5 km / 0.9 miles of roadway will be fully rehabilitated, with nearly 15,000 m² / 161,500 sq ft of surface area to be treated.

Technical Description of the Work

Preparatory phase

- additional geotechnical studies;
- assessment of existing underground utilities;
- installation of the construction-site base;
- temporary signage and detours;
- communication with nearby residents and business owners.

Roadwork

- removal and disposal of old pavement layers;
- reinforcement of the roadbed;
- installation of a new low-noise asphalt surface;
- raised-table treatments at intersections;
- creation of 20 mph zones and shared streets limited to about 12 mph;
- installation of speed humps at strategic points.

Pedestrian and bicycle improvements

- sidewalk widening to a minimum of 1.80 meters / about 6 feet;
- creation of protected bike lanes, 1.50 meters / about 5 feet wide;
- installation of bike racks at strategic locations;
- raised crosswalks with tactile warning surfaces;
- removal of architectural barriers.

Utilities and stormwater management

- rehabilitation of the stormwater drainage network;
- installation of landscaped bioswales for alternative stormwater management;
- new-generation inspection chambers and storm drains;
- integration of runoff-water filtration systems.

Landscaping and street furniture

- planting of 35 street trees suited to urban conditions;
- installation of 12 integrated planters;
- placement of 28 ergonomic and inclusive public benches;
- installation of 40 recycling and waste bins;
- installation of 15 drinking-water fountains;
- complete renewal of street lighting, coordinated with the Public Lighting Modernization Project.

Innovation and Sustainable Approach

- use of at least 30% recycled asphalt mixes;
- use of permeable concrete in selected pedestrian areas;
- integration of bio-based materials for street furniture;
- creation of “garden streets” with strong vegetation coverage;

- rainwater harvesting for irrigation;
- urban sensors to monitor air quality and noise.

Phasing and Preliminary Schedule

The project is structured around four main phases:

- Phase 1 — September to November 2023: Republic Avenue, north section; detailed diagnostics and preparatory work.
- Phase 2 — December 2023 to February 2024: Republic Avenue, south section; City Hall Square.
- Phase 3 — March to May 2024: Victor Hugo Street; intersection with School Street.
- Phase 4 — June to August 2024: Commerce Street; finishing work and landscaping.

Final acceptance of the work is scheduled for September 2024, for a total duration of 12 months.

Detailed Budget

Budget item	Amount	Share
Preliminary studies and design services	\$60,000	8%
Roadwork and structural layers	\$375,000	50%
Drainage and sewer-related networks	\$112,500	15%
Bicycle and pedestrian facilities	\$90,000	12%
Street furniture and lighting	\$67,500	9%
Planting and green spaces	\$45,000	6%
Total	\$750,000	100%

Organization and Governance

The project is led by Willow Creek Public Works in collaboration with:

- an urban architecture firm acting as lead consultant;
- a civil engineering firm specializing in roads and utilities;
- a landscape designer;
- an urban mobility expert.

A steering committee meets monthly and includes the elected official responsible for planning and public works, the town administrator, the public works director, downtown business representatives, a neighborhood association representative, and a representative of people with reduced mobility.

Communication and Public Engagement Plan

- quarterly public meetings with residents;
- a weekly information desk about construction activity;
- a dedicated website with project updates;
- a mobile app for reporting construction-related issues;
- on-site information panels with interactive QR codes;
- a monthly construction newsletter distributed to nearby residents;
- educational site visits for schools and youth programs.

Impact on Local Life and Mitigation Measures

- work by limited sections of no more than 150 meters / about 500 feet;
- systematic maintenance of access to businesses and homes;
- safe temporary pedestrian routes;
- restrictions on noisy work hours;
- a compensation mechanism for businesses severely affected;
- a shared delivery service during construction periods.

Expected Benefits

- 30% reduction in downtown traffic accidents;
- 25% increase in pedestrian traffic;
- estimated 15% increase in business revenue after renovation;
- 2 to 3°C reduction in urban heat-island intensity during summer;
- 4 to 6 dB reduction in traffic noise;
- improved air quality through reduced vehicle traffic;
- a more pleasant living environment and stronger property values;
- reinforced urban identity and sense of belonging.

Evaluation and Monitoring Indicators

- adherence to schedule and budget;
- satisfaction surveys among users and business owners;
- before-and-after pedestrian and cyclist counts;
- air-quality and noise measurements;
- changes in commercial vacancy rates;
- number of events held in the renovated public space;
- reduction in road-maintenance interventions over the medium term.

Future Outlook

This renovation will serve as a pilot project for future interventions on Willow Creek's local road network. The innovations tested and the lessons learned will help refine the town's approach in other neighborhoods. The project supports a gradual transition toward a more sustainable, accessible, and pleasant town.

Conclusion

The central road renovation is much more than a technical resurfacing effort. It is a transformative project that will restore the heart of Willow Creek to a level of quality, functionality, and appeal that meets current needs while preparing for the future. Through its integrated approach, it addresses modern urban challenges: sustainable mobility, universal accessibility, climate resilience, and quality of life. The investment made today will benefit the town for decades to come.